

BSR Transition Risk — Methodology

Reflects the code as built through P0–P3. This document specifies the flow, mathematics, and logic of the standalone transition-risk engine (`engine/transition/`) and app (`transition_app/`). It is generated to be marked up: each section states what the code does, the exact formula, the data behind it, and the judgment parameters you can challenge.

Companion to the physical-risk methodology. Screening-grade throughout — not a regulatory disclosure without specialist review.

1. Overview

The engine decomposes the financial impact of decarbonisation on an asset or portfolio into **four channels**, each mapped to one **TCFD transition-risk category**, quantified by one **engine layer**, and routed to exactly one **financial destination** (the *non-duplication* rule):

TCFD category	Layer	Mechanism	Routes to
Policy & Legal	L1	(Scope 1+2) × NGFS carbon price × (1 – pass-through)	Cash-flow OpEx
Technology	L2	Wright's-Law cost crossover / demand collapse → logistic impairment	Cash flow + asset value
Market	L3	Sector shock → Leontief inverse → focal input cost	Cash-flow input cost
Reputation	L4	Sautner CCExposure (z-scored) × per-SD elasticities	WACC (default) or cash flow

On top of the four layers sit a **decision & assurance layer**: Monte-Carlo uncertainty, tornado sensitivity, PCAF financed emissions, Implied Temperature Rise, PACTA-style alignment, peer benchmarking, marginal-abatement-cost economics, a budget optimizer, and an IFRS S2 / ESRS E1 disclosure export.

How to run

```
pip install streamlit pandas numpy plotly openpyxl
streamlit run transition_app/Home.py          # the app
python -m pytest tests/test_transition.py -q   # the tests (all green)
```

The engine + tests need only numpy/pandas/openpyxl/pytest. `pyam` and `pymrio` are only for the data-refresh scripts (NGFS prices, EXIOBASE MRIO).

2. Architecture & module map

engine/transition/	
data_loader.py	cached JSON loaders; ISO3→NGFS region; scenario→NGFS mapping
carbon_pricing.py	L1 – carbon cost, pass-through, abatement pathway, priced fraction
learning_curves.py	L2 – Wright's law, Lafond bands, crossover, logistic impairment
adaptive_capacity.py	L2 modifier – ambition / positioning / capture / transition capex (residual)
network_propagation.py	L3 – world 20×20 Leontief propagation
network_mrio.py	L3 – 20×49 EXIOBASE MRIO + Reisch endogenous-default cascade
cc_exposure.py	L4 – z-standardised CCExposure × per-SD elasticities + routing
transition_engine.py	orchestrator: run_asset_transition / run_portfolio_transition
transition_dcf.py	combined physical+transition DCF hook
uncertainty.py	Monte-Carlo over price / pass-through / elasticity
sensitivity.py	one-at-a-time tornado
alignment.py	financed emissions, ITR, pathway alignment, peer benchmark
macc.py	marginal abatement cost curves (decarbonise vs pay)
optimizer.py	budget-constrained merit-order abatement allocation
levers.py	decarbonization lever library – sector→lever map + plan overlay (reference)
data/transition/	
carbon_prices_ngfs.json	NGFS Phase V REMIND-MAGPIE (real, ×1.18 → USD2020)
sector_pass_through.json	sector-median pass-through
learning_curves.json	learning rates + LCOE base costs (IRENA 2024 / Lazard 2025)
sector_pathways.json	demand pathways (production index, 2025=1.0)
cc_exposure_proxy.json	sector CCExposure (real ×10 ³ scale) + pooled z-base + elasticities
io_matrix.json	20×20 world direct-requirements matrix (EXIOBASE world totals)
io_matrix_mrio.npz + meta	20×49 EXIOBASE MRIO (980×980)
sector_taxonomy.json	30 sectors (20 core + 10 io_proxy): tech pairs, fossil flag, emissions
macc.json	per-sector abatement measures (AR6 WG3 / IEA-informed)
lever_library.json	29 decarbonization levers × 5 domains + sector→lever value-chain
adaptive_capacity.json	scenario→ambition, lever→readiness, sector pivot-capex ratios, regional
region_factors.json	geographic resource/cost zones → region-aware LCOE (renewable / fossil)

Execution flow

```
① Data Entry → intake per entity: sector, Scope 1/2/3, replacement value, revenue,
               target_year, priced_pct, attribution_pct
               |
               ▼
run_portfolio_transition(assets, scenarios, ...settings...)
  | for each (scenario, asset):
  |   L1 carbon_cost_timeline(...) → net_carbon_opex[y]
  |   L2 compute_stranding(...) → impairment[y], revenue_erosion[y]
  |   L3 propagate (world | mrio+cascade) → indirect_input_cost[y]
  |   L4 compute_exposure_premium(...) → revenue_modifier[y] OR wacc_bps
  |   total_cf[y] = L1 + L2_rev + L3 + L4 ; impairment separate
  |
  ▼
Results / category pages / Alignment / Abatement / Audit
      → PV, Monte-Carlo P5–P95, tornado, ITR, PCAF, MACC, XLSX + IFRS S2 export
```

3. Layer 1 — Policy & Legal (direct carbon cost)

Module: `carbon_pricing.py`. For each (scenario, region, year, sector):

3.1 Carbon price

`get_carbon_price(scenario, year, band)` linearly interpolates the NGFS trajectory and holds it flat outside 2025–2050, then multiplies by a Monte-Carlo perturbation `price_scale` (1.0 = base):

```
price(y) = interp(NGFS[scenario][band], y) × price_scale
```

Region **band** ∈ {advanced, emerging, rest_of_world} from `get_ngfs_region(IS03)`. IPCC SSP scenarios map to their closest NGFS analog.

3.2 Abatement pathway (P0)

An entity with a Scope 1+2 net-zero **target_year** decarbonises linearly; carbon cost then applies only to the declining residual:

```
abatement_mult(y) = 1.0                                if no target or y ≤ 2025
                   = residual                            if y ≥ target_year          (residual = re
                   = 1 + (y-2025)/(target-2025)·(residual-1) otherwise
```

3.3 Cost decomposition

```
direct      = (Scope1 + Scope2) × abatement_mult(y)
pass_through = min(1, sector_PT × pt_scale)           (or per-asset override)
opportunity  = direct × price                          full carbon opportunity cost
passed_through = opportunity × pass_through           ← L3 market shock (undiminished by free al
net_compliance = opportunity × priced_fraction       allowances the firm actually buys
absorbed     = net_compliance – passed_through       firm's net margin impact (windfall if < 0)
scope3_indirect = Scope3 × price × (1 – pass_through)
net_carbon_opex = absorbed + scope3_indirect         ← what hits the firm's cash flow
```

- **Windfall economics (Round-1 P5).** The marginal carbon price sets the pass-through opportunity cost regardless of free allocation (Sijm 2012), so pass-through — and the L3 shock — are based on the **full opportunity cost**; `priced_fraction` reduces only the firm's own compliance leg. With generous free allocation and high pass-through, `absorbed` goes **negative** (a windfall gain). At the default `priced_fraction = 1` this reduces to `absorbed = opportunity × (1 – PT)` (legacy).
- **priced_fraction** models free allocation / partial ETS coverage (P0). Default 1.0.
- **Scope-2 & Scope-3 incidence (external review).** Purchased-electricity carbon (Scope 2) and upstream Scope 3 are *supplier* costs that reach the firm through prices, which Layer 3 already models. Charging them again as a direct L1 liability double-counts, so **scope2_mode** and **scope3_mode** now default to **auto**: both are dropped from L1 whenever Layer 3 is enabled — Scope 2 unless the firm pays an explicit carbon charge on its electricity (`scope2_mode="direct"`). `full` keeps them in L1 and should be used only with Layer 3 off. `scope3_incidence` (R3) lets the analyst set the supplier-to-buyer pass-through directly.

Data: `carbon_prices_ngfs.json` — real **NGFS Phase V** (released Nov 2024) REMIND-MAGPIE 3.3-4.8 prices pulled live from the IIASA `ngfs_phase_5` explorer, rebased ×1.18 (US GDP deflator 2010 → 2020) to ~USD2020. Six scenarios carry real data; `divergent_net_zero` + three IEA scenarios retain Appendix-D placeholders. (Round-1 P4: the Kotz et al. 2024 retraction affects only NGFS physical-damage variables, not the transition/price pathways — L1 uses standard REMIND-MAGPIE transition price variables and is unaffected.) **Pass-through:** `sector_pass_through.json`, sector medians (Sijm 2012; Fabra & Reguant 2014; Cludius 2020).

4. Layer 2 — Technology (learning curves & stranding)

Module: `learning_curves.py`. The projection and its uncertainty band implement the **Farmer–Lafond** empirically-grounded experience-curve forecast (Farmer & Lafond 2016; Way, Ives, Mealy & Farmer 2022): a geometric random walk with drift on log-cost, whose forecast error grows as $\sigma \cdot \sqrt{(\text{horizon})}$. Way et al. (2022)'s central finding — clean tech on persistent exponential decline while fossil stays flat, so a **fast transition is the low-cost path** — grounds both the crossover logic here and the scenario-narrative ambition defaults in §4.6. Learning rates and base costs are from primary datasets (IRENA 2024, Lazard 2025 v18, the Oxford/Way 2022 dataset, BNEF 2024, Ziegler & Trancik 2021) — **not** the BSR lever library.

4.1 Wright's Law

$$\text{Cost}(Q) = \text{Cost}_0 \times (Q/Q_0)^b, \quad b = \log_2(1 - \text{LR})$$
$$Q/Q_0 = (1 + g)^{(y - 2025)}, \quad g = \max(0, \text{scenario deployment growth})$$

Negative growth (demand decline) is clamped to 0 — cumulative capacity is monotonic; demand decline is handled by the sector pathway, not the cost curve.

4.2 Farmer–Lafond distributional band

$$\sigma_t = \text{lafond}_\sigma \times \sqrt{(y - 2025)}$$
$$\text{cost_lo} = \text{pred} \times e^{(-\sigma_t)}, \quad \text{cost_hi} = \text{pred} \times e^{(+\sigma_t)}$$

4.3 Stranding trigger (either fires)

- **Cost crossover** — first year challenger cost $\leq$ incumbent cost.
- **Demand collapse** (fossil-dependent sectors only) — first year the sector pathway falls below 0.5.
- **Tech-substitution stranding (diagnostic fix #3)**. Even where DEMAND holds (steel, cement), a cost crossover obsoletes the incumbent asset (blast furnace, wet kiln) and it is written off. The carbon-specific share (`TECH_SUBSTITUTION_SHARE` ≈ 0.45) strands in proportion to how much the firm transitions (ambition), paired with the capex that builds the replacement. `Cap = max(demand_cap, tech_cap) × base`. Steel now shows ~\$4B stranding on a \$18B works (was \$0); cement/chemicals stay ~\$0 (no cost crossover — they need costly CCUS, not obsolescence by a cheaper tech).
- **P6 — carbon-inclusive crossover (opt-in)**. When enabled, the incumbent's effective cost adds its carbon cost `carbon_price(scenario, region, y) × carbon_ef_per_unit` (Scope-1 emission factor in the tech's cost unit; fossil incumbents only). A rising carbon price then pulls the crossover — and stranding — earlier. Off by default → pure-LCOE crossover, preserving anchors.
- **R6 — crossover band**. The Lafond $\pm 1\sigma$ bands give an *early* (challenger-low vs incumbent-high) and *late* (challenger-high vs incumbent-low) crossover year that bracket the central estimate; surfaced as a crossover range in the Technology page and as an impairment tornado driver.

4.4 Impairment

$$\text{frac}(y) = 1 / (1 + e^{(-\text{slope} \cdot (y - \text{crossover}))}) \quad \text{logistic}$$
$$\text{slope} = \text{per-sector (R7; default 0.20)} \quad \text{e.g. power_coal 0.20 (anchor), ICE auto}$$
$$\text{base} = \text{replacement_value} \quad \text{if fossil-dependent}$$
$$= \text{replacement_value} \times f \quad \text{otherwise, } f = \text{non_fossil_base_fraction (R1; default 0.5)}$$
$$\text{cap} = \max(0, 1 - \text{pathway}(2050)) \times \text{base} \quad \text{scenario-scaled ceiling}$$
$$\text{annual_impairment}(y) = (\text{frac}(y) - \text{frac}(y-1)) \times \text{cap}$$

Impairment is a **balance-sheet** figure, reported separately from cash flow. **% stranded = recognised, not the ceiling (external review finding 1)**. The headline `stranded_fraction_2050` is the *recognised* cumulative impairment over 2025–50 ÷ replacement value, so it always equals the dollar impairment (e.g. coal $\approx 53\% = \$266\text{M} \div \500M). The logistic *level* at 2050 (which includes pre-2025 stranding never

booked) is reported separately as the `strandable_ceiling` (coal $\approx$ 98%). The two must not be conflated. **R7 — per-sector slope** is read from `sector_taxonomy.stranding_slope` (calibrated to asset-turnover speed; power fast, heavy industry & networks slow); the two documented calibration anchors (power_coal, oil_refining) are held at the legacy 0.20. **R1 — non-fossil base fraction** is configurable in the UI.

4.5 Revenue erosion

```
revenue_index(y) = sector_pathway(y)           (production index, 2025 = 1.0)
revenue_erosion(y) = max(0, revenue × (1 - index(y))) ← cash-flow cost
```

Data (P0 refresh): power-sector base costs are **LCOE (USD/MWh)** from **IRENA Renewable Power Generation Costs in 2024** and **Lazard LCOE+ 2025 v18** (coal 118, gas 76, solar 43, onshore wind 34, storage 170). Learning rates: Way et al. 2022 / IRENA. Industrial base costs remain sector estimates.

4.5b Product use-phase risk (stress-test fix #2)

A maker of carbon-emitting **products** — diesel machinery, engines, ICE vehicles, fuels — carries a risk its own Scope 1+2 (own operations) completely misses: as its *customers* face carbon prices and switch to cleaner alternatives, demand and margin on the carbon-emitting product line fall. This is the **use-of-sold-products** channel (GHG Protocol Scope 3 category 11), entered separately as `scope3_use_phase_tco2`:

```
product_use_phase_cost(y) = use_phase_scope3 × carbon_price[y] × USE_PHASE_INCIDENCE × (1 - capt
```

with `USE_PHASE_INCIDENCE = 0.15` — the fraction of the customers' carbon burden that feeds back to the maker as lost demand/margin (deliberately conservative, and distinct from the sector *volume* pathway of §4.5, which captures the quantity effect; the use-phase term is the residual price/margin pressure). `(1 - capture)` credits a maker that pivots its product line to clean alternatives. Routed once, as a cash-flow cost — distinct from **upstream** Scope 3 (in L3) and own-operations Scope 1+2 (in L1). For Caterpillar this term ($\approx$ 360 Mt use-phase) is the single largest cost, correctly reframing an asset-light equipment maker whose real transition risk is that its diesel product line strands.

4.6 Adaptive capacity — gross exposure → residual risk

Module: `adaptive_capacity.py` + `adaptive_capacity.json`. §4.5 as written erodes $\sim$ 100% of the incumbent product's revenue — a *frozen* company that captures nothing from the low-carbon business (an ICE automaker losing all revenue with \$0 from EVs). That is **gross vulnerability**, not residual risk. This layer (**ON by default**) converts it into "how the company looks if it follows the scenario pathway, given where it starts." Approach follows BSR's *Decarbonization Lever Library: Mapped Sectoral Transition Pathways* (Nov 2025) — business-transformation / systemic-adaptability framing.

Three levers:

ambition $A \in [0,1]$ default from the SCENARIO NARRATIVE (NZ-2050 ≈ 0.90 , Current Policies ≈ 0.60)
 overridable. Also implies an emissions pathway (residual = $1-A$) that lowers emissions when no explicit decarbonisation target is set – so the pivot is never 100%
 positioning $P \in [0,1]$ "science": $0.35 \cdot \text{plan_strength}(\text{target} + \text{in-plan lever coverage}) + 0.30 \cdot \text{sector}$
 $\text{lever readiness (Lever-Library maturities)} + 0.20 \cdot \text{emissions-vs-sector}$
 $+ 0.15 \cdot \text{plan coverage}$. "art": a manual override on Data Entry.
 $\text{capture} = \min(A \cdot P^{1.15}, \text{opportunity_ceiling})$ share of gross erosion offset by pivoting

Effects (each routes once):

$\text{net_revenue_erosion}(y) = \text{gross_erosion}(y) \times (1 - \text{capture})$ ← CASH FLOW (reduced)
 $\text{transition_capex}(y) = \text{phased}[A \cdot \text{replacement_value} \cdot \text{sector_pivot_ratio} \cdot \text{region_multiplier} \cdot \text{positioning_capex_factor}]$ ← CASH FLOW (increased)
 $\text{strandable_base} \times = (1 - \text{already_transitioned}), \text{ already} = 0.5 \cdot P$ ← BARRIERS TO ENTRY

Transition capex is company-provided (they know their plan), else estimated from pivot scale $\times$ sector ratio $\times$ a **geographic/context buffer** (advanced 1.0, emerging 1.1, RoW 1.2) $\times$ a positioning factor (laggards pay up to 1.6 $\times$, movers 0.8 $\times$). Worked example — automaker (\$100B rev) under NZ-2050: gross \$791B PV → **\$318B** if strongly positioned & ambitious (80% capture, \$14B capex, less stranding), vs **\$740B** if a poorly-positioned laggard (10% capture, \$24B capex). Present positioning drives the outcome. Positioning derivation is a screening heuristic — the "art" override exists for exactly the cases it cannot cleanly quantify.

4.7 Geography — region-aware technology costs

Data: `region_factors.json`. The learning-curve LCOEs are world averages, but a watt in Texas or MENA is far cheaper than in Northern Europe or Japan, and green steel / ammonia / cement is cheapest where clean power is cheap. Each ISO3 maps to a **resource zone** carrying three multipliers on a technology's LCOE before the crossover comparison:

clean power (renewable_lcoe_factor) → solar, wind, storage, PPAs
 green H2 etc. (green_h2_factor) → electrolyser, H2-DRI steel, green ammonia, SAF, biorefinery
 fossil (fossil_lcoe_factor) → REGIONALLY-PRICED energy only: natural gas + grid electricity

`_project_cost` multiplies by the zone factor, so `find_crossover_year` and stranding are **region-aware**. Only regionally-priced fossil energy (gas, grid power) carries a factor; globally-traded commodities (coking coal, crude oil, and the coal/oil-based incumbents — blast furnace, cement, refining, jet/bunker fuel) stay at 1.0, so the **green challenger's** regional cost drives the crossover — green steel is earliest where clean power / green H₂ is cheapest (the intuitive result). Factors follow IRENA regional LCOE dispersion (best solar ~\$0.03/kWh in MENA/sunbelt/Chile/Australia/Iberia vs ~\$0.06–0.09 in N. Europe/Japan) and regional gas prices; the USA-national / global-average zone is the 1.0 reference, so the world-average curves and worked-example anchors are unchanged.

Sub-national resolution (US & Europe — highest-importance markets). Enter the region as an ISO3-prefixed code — `USA-TX`, `USA-CA`, `USA-SW`, `ESP-S`, `ITA-N` ... — resolved sub-national → country → global, and stripped to the country for the carbon-price band. US grid/resource zones: Texas (ERCOT),

Southwest, Midwest wind belt, California, Southeast, Northeast, Pacific NW. Europe is country-resolved into UK & Ireland, N. Europe, Nordics, Iberia, S. Europe, C./E. Europe. Effect — **green-steel (H2-DRI) crossover: US Southwest 2026 · Texas 2027 · California 2034 · US-national 2036 · US Northeast 2039; Spain 2031 · UK 2038 · Germany 2040 · Japan 2042** (same plant). A watt in Texas ≠ one in California; green steel in Iberia ≠ in Germany. *Limitation*: US is resolved to ~7 grid regions and Europe to country/macro-region — not to individual states/provinces or specific balancing authorities yet.

5. Layer 3 — Market (production-network propagation)

Modules: `network_propagation.py` (world), `network_mrio.py` (MRIO + cascade).

5.1 Sectoral shock

Each sector's carbon shock as a fraction of its output:

$$s_i = \min(1.0, \text{emission_intensity}_i \times \text{price} \times \text{pass_through}_i \times 1e-6)$$

(emission_intensity in tCO₂ per **M\$ of sector gross output**; 1e-6 → per-USD.) **Scale correction (external review finding 3)**: the intensities were ~1000× too low (implied t/\$k), which made L3 invisible; they are now realistic EEIO/EXIOBASE-informed direct Scope-1 intensities (fossil sectors O(1000s) tCO₂/M\$, services O(10s)), verified against the fixtures by a unit test. Each sector shock is **clamped to 1.0** — carbon cost cannot inflate a sector's output price by more than 100% in the linear Leontief basis; beyond that the sector is stranding (Layer 2), not passing input cost. The shock also responds to the Monte-Carlo `price_scale` and `pass_through_scale` in both the world and MRIO paths. Using sector-typical exposure avoids double-counting the focal asset's own pass-through.

5.2 Leontief propagation

$$\begin{aligned} L &= (I - A)^{-1} \\ \text{total_shock}_j &= \sum_i L[i,j] \cdot s_i \\ \text{propagated}_j &= \text{total_shock}_j - \text{own} \quad (\text{own} = s_j, \text{ the single L1 direct round} - P3) \\ \text{indirect_cost} &= \text{propagated}_j \times (\text{asset_revenue} \times \text{intermediate_input_share}) \times \text{absorption} \end{aligned}$$

Input-share basis (diagnostic fix #1). L3 scales by the firm's carbon-EXPOSED input base — `revenue × intermediate_input_share` (bought-in intermediate inputs, 1 – value-added) — NOT total revenue. Previously a labour/margin-heavy firm (financial services 96%-of-cost L3, ~\$7B on a bank) booked an implausible supply-chain carbon cost; with the input-share basis (finance ≈ 0.28, refining ≈ 0.85) that collapses to ~\$2B and heavy processors keep a large, legitimate L3. **CES damping**: off-diagonal A scaled by 1/σ before inversion (σ = substitution elasticity; Papageorgiou 2017 range 1.3–3.0). σ=1 is Cobb-Douglas.

R2 — partial pass-through (opt-in). `absorption` is the share of the propagated upstream cost the focal firm *absorbs* rather than passing to its own customers. Default 1.0 (full absorption, legacy). When enabled it is set to `1 – pass_through_j`, so sectors with pricing power recover part of the input-cost shock downstream and their net margin impact falls.

5.2b Report-anchored L3 (stress-test fix #1)

When a firm **discloses its own upstream Scope 3** (GHG Protocol categories 1–9), the model prices that reported quantity directly instead of estimating it from a sector-typical Leontief propagation:

```
indirect_cost[y] = reported_upstream_scope3 × carbon_price[y] × L3_SCOPE3_INCIDENCE × absorptior
```

with `L3_SCOPE3_INCIDENCE = 0.5` (overridable via `scope3_incidence`) — the share of the upstream carbon bill that reaches the focal firm through supplier prices rather than being absorbed further up the chain. This makes the value-chain cost track the company's *actual* disclosed footprint (e.g. BASF 90 Mt, Nike 9.5 Mt) rather than a generic estimate. The Leontief path of §5.2 remains the fallback for **non-disclosers** (`scope3 = 0`) and is bypassed only in `auto` mode (the default, where the upstream Scope 3 is *not* already in L1). Because the disclosed figure is a firm-level total, the report-anchored path does not produce a modelled upstream **sector decomposition** — the "top upstream sources" drill-down is shown only for Leontief-estimated firms.

5.3 Resolution (P2)

- **World** (default): 20×20 single-region matrix (EXIOBASE-3 world totals, `io_matrix.json`).
- **Multi-region**: full **20 sectors × 49 regions = 980×980** EXIOBASE MRIO (`io_matrix_mrio.npz`). Cross-border supply chains are explicit and each region's sectors pay carbon on **that region's price band**, so an EU refiner's indirect cost reflects emerging-market upstream at a different carbon price. Leontief inverts in ~0.1 s.

5.4 Endogenous-default cascade (P2, Reisch et al. 2025)

Beyond linear Leontief, nodes absorbing input-cost shock above threshold θ pass an amplified shock downstream, iterated to convergence:

```
s_eff = s
repeat: total = L · s_eff ; over = max(0, total -  $\theta$ ) ; s_eff = s + contagion · over
```

Default off; triggers only for the most-exposed nodes under stress ($\theta \approx 0.02 \rightarrow \sim 1.25\times$ for CHN steel; $\theta = 0.01 \rightarrow \sim 6.8\times$).

6. Layer 4 — Reputation (CCExposure, z-standardised)

Module: `cc_exposure.py`. **P0 fix (D3):** raw 0–2 proxies inflated elasticities ~500–1000×; the engine now consumes **z-scores** against the real Sautner distribution and prices **per standard deviation**.

```

rawx = sector_exposurex × scenario_modifierx           x ∈ {opp, reg, phy}
zx   = (rawx - pooled_meanx) / pooled_sdx
z_total = (Σ rawx - pooled_mean_total) / pooled_sd_total

credit_spread_bps = 12·z_reg + 6·z_phy
equity_premium_bps = 18·z_reg + 9·z_phy - 25·z_opp      (downside raises cost of equity; opportunity discount)
revenue_growth_bps = 35·z_opp - 25·z_reg

```

Sign fix (trial-run finding). The equity premium previously used $50 \cdot z_{total}$, which lumps opportunity in with downside — so a renewables firm (high opportunity) got a *higher* cost of capital for being a climate winner, and the WACC and cash-flow routes disagreed in sign. It now prices the **downside** components (regulatory + physical) positively and the **opportunity** component as a **discount**, consistent with the carbon-premium / greenium literature (Bolton & Kacperczyk 2021; Zerbib 2019). Result: renewables ≈ **-33 bps** ΔWACC (cheaper capital), coal ≈ **+27 bps** (penalty), neutral sectors ≈ 0. The **pooled distribution** is Sautner et al. (JoF 2023) **Table 1** (firm-year, ×10³): opp 0.391/1.344, reg 0.049/0.264, phy 0.013/0.103, total 0.943/2.443. **Sector exposures** are anchored to Sautner **Table 4** by SIC industry where available (utilities → power, petroleum refining, transport equipment → autos, primary metal → steel ...), estimated from adjacent industries otherwise.

Effect: average-exposure sectors carry ≈0 premium; coal keeps ~29 bps credit and a **+27 bps WACC penalty**; renewables now receive a **-33 bps WACC discount** (and, on the cash-flow route, a revenue uplift) — a climate winner is financed more cheaply, not penalised.

Routing (non-duplication): `ROUTE_WACC` (**default, R5**) adds the equity premium to the discount rate and zeroes the revenue modifier; `ROUTE_CASHFLOWS` applies the revenue-growth modifier to cash flows instead. **Firm override (P2):** a licensed firm-level CCExposure feed replaces the sector median via `firm_override`.

R5 — elasticity provenance (which coefficient is real). Only one of the three L4 elasticities is sourced, and the default routing reflects that:

Coefficient	Value	Provenance	Route
equity → WACC	18·z_reg + 9·z_phy - 25·z_opp bps	SOURCED (downside) — Sautner Pricing: cost-of-capital premium concentrated in regulatory/physical exposure; opportunity discount informed by the greenium literature	WACC (default)
credit spread	12·z_reg + 6·z_phy bps	UNSOURCED placeholder — plausible sign/magnitude, not fitted	diagnostic only
revenue growth	35·z_opp - 25·z_reg bps	UNSOURCED — market-opportunity judgement, not an estimated elasticity	manual cash-flow overlay

Because the equity → WACC channel is the only empirically grounded one, `ROUTE_WACC` is the default and the cash-flow revenue route is presented in the UI as a **manual analyst overlay**, not a model output. The credit-spread figures are shown for diagnostics but never routed into headline numbers.

7. Orchestration & non-duplication

`run_asset_transition(asset, scenario, ...)` composes the layers. Each channel routes once:

```
total_cf(y) = L1 net_carbon_opex
             + L2 revenue_erosion
             + L3 indirect_input_cost
             + L4 revenue_modifier (as a cost: -modifier)    (only if ROUTE_CASHFLOWS)
impairment(y) = L2 annual_impairment    (balance-sheet, NOT in total_cf)
ΔWACC_bps = equity_weight·equity_premium + debt_weight·credit_spread·(1-tax)    (only if ROUTE_V
```

Capital-structure-weighted WACC (external review finding 4). The L4 premium is no longer `credit + equity` added one-for-one; the equity premium enters the cost of equity and the credit spread the after-tax cost of debt, weighted by capital structure (default 60/40, 25% tax). It is now **actually applied** in `transition_dcf.compute_combined_dcf` (added to `climate_risk_premium` for the transition and combined DCFs) — previously it was computed but never discounted, so L4 → WACC had no valuation effect. `run_portfolio_transition` runs all assets × scenarios and returns `{scenario: [TransitionAssetResult]}`, each tagged with a **data-quality flag** (firm / sector-proxy / degraded) that surfaces on the Audit page.

Firm-level roll-up (diagnostic fix #4). Assets sharing a `firm_id` roll up into one diversified company via `firm_rollup()` — a multi-line, multi-region firm gets one consolidated cost/impairment view across its business lines (surfaced on the Results page). Each line keeps its own sector- and region-specific positioning (correct: a steel division and a data-centre division transition differently). This is an INDEPENDENT sum — group-level correlation, cross-subsidy, shared capital and a single optimised group plan are **not** modelled (documented limitation).

Present value & discount basis (Round-1 P2 + external review). Carbon prices are **real** (USD2020), so PV discounts the real cash flows at a **real** rate = nominal WACC – long-run inflation. Discounting real flows at the nominal WACC would systematically understate PV. Both are app inputs (nominal WACC default 9%, inflation 2.5% → real 6.5%). **One timing convention** is now used everywhere — end-of-year $PV = \sum cost / (1 + real)^{(y - 2025 + 1)}$ — across the DCF engine, Monte-Carlo, tornado, disclosure export and the stranded-impairment PV (previously these mixed `y-2025` and `y-2025+1`).

Impairment vs cash-flow cost are NOT additive (Round-1 P1). L2 stranded impairment is the PV writedown of the same future cash flows whose erosion already feeds the cash-flow cost. They are two **lenses on one loss** — reported side by side, never summed into a combined total (enforced in the dashboard, XLSX, and disclosure export; guarded by a test).

8. Uncertainty & sensitivity

8.1 Monte-Carlo (`uncertainty.py`)

Re-runs the four layers over N draws (default 400), perturbing:

price_scale	~ LogNormal(0, 0.20)	carbon-price / policy path
pass_through_scale	~ Normal(1, 0.10)	cost incidence
elasticity σ	~ Uniform(1.0, 2.5)	network substitution

Returns P5 / P50 / P95 of PV transition cost and stranded impairment. Seed-deterministic.

U1 — the P5–P95 range is a *conditional floor*, not full uncertainty. It is conditional on the selected scenario and varies only the three sampled parameters — it excludes scenario/policy-path uncertainty, the L2 stranding trigger-year and the demand pathway. The UI labels the P95 accordingly and points to the tornado for the excluded drivers.

8.2 Tornado (`sensitivity.py`)

One-at-a-time swing of each driver, all others at base, sorted by |swing|. Two targets (U1): - **Cash-flow cost** — carbon price $\pm 30\%$, pass-through $\pm 20\%$, L3 input-cost pass-through (R2), σ 1.0–2.5, Scope-3 mode, discount $\pm 2\text{pp}$. - **Stranded impairment** — crossover trigger (pure-LCOE vs carbon-inclusive, R6/P6), stranding slope 0.12–0.35 (R7), non-fossil base share 0.25–0.75 (R1), carbon price $\pm 30\%$, discount $\pm 2\text{pp}$. This view surfaces the trigger-year and base assumptions that dominate stranding but are invisible to the cash-flow tornado and to the Monte-Carlo floor.

9. Decision & alignment layer

9.1 Financed emissions — PCAF (`alignment.py`)

Attributed emissions = $\sum \text{attribution_pct} \times (\text{Scope } 1/2/3)$. 100% = corporate own-asset view; a lender/investor enters their stake.

9.2 Emissions-budget temperature score (screening ITR proxy)

A **screening indicator**, NOT a standard-compliant Implied Temperature Rise (SBTi / CDP-WWF temperature scoring). Each entity's abated Scope 1+2 pathway vs a 1.5 °C linear-to-net-zero budget:

budget = $0.5 \times E_0 \times (2050 - 2025)$	area under a straight line to zero
ratio = $\text{cumulative_actual} / \text{budget} - 1$	
score = $\text{clamp}[1.5, 4.0](1.5 + 1.2 \times \text{ratio})$	floored at 1.5 °C best case
portfolio score = $\text{Scope}_{1+2} \times \text{attribution weighted mean}$	

Net-zero-by-2050 ≈ 1.5 °C; flat emissions ≈ 2.8 °C. **External review:** the old 1.2 °C lower bound implied sub-1.5 alignment and was not defensible — the floor is now the 1.5 °C best case. Report this as a screening score, not a certified ITR.

9.3 Pathway alignment (PACTA-style)

Compares the entity's Scope 1+2 decline by 2050 to the scenario's sector demand pathway; gap ≤ 0.05
 → aligned, ≤ 0.25 → lagging, else misaligned.

9.4 Peer benchmarking

Entity economic intensity (tCO₂ Scope 1+2 / \$M revenue) vs the **real Sautner firm-level Carbon Intensity distribution** (JoF 2023 Table 1: median 11, p75 84.6 across ~10k firms) → quartile position.

9.5 Marginal abatement cost curves (`macc.py`)

Per-sector measures (cost USD/tCO₂, potential share of Scope 1+2). At carbon price P, all measures with cost ≤ P are cost-effective:

```
abated_frac = Σ potential (cost ≤ P) , capped at 1
abatement_cost = Σ cost × potential × emissions
carbon_cost_avoided = abated × P , net_benefit = avoided - cost
```

9.6 Budget optimizer (`optimizer.py`)

Merit-order greedy: sort all measures across the portfolio by \$/tCO₂; fund no-regret (negative-cost) measures always, then buy cheapest-first until the budget is exhausted. Returns abated tonnes, net spend, marginal-cost frontier, residual carbon cost, and per-asset allocation. Optimal for max tonnes per dollar under a linear MACC.

9.7 Disclosure export

`build_disclosure_report` generates a Governance → Strategy → Risk Management → Metrics report mapped to **IFRS S2** clauses and **ESRS E1** datapoints, plus a provenance-stamped multi-sheet XLSX. **U2 — the experimental endogenous-default cascade (\$5.4) is hard-excluded from the disclosure in code:** the report recomputes its figures with `cascade=False` regardless of the UI toggle, and states the exclusion, so a research amplification can never leak into a regulatory-style export.

9.8 Decarbonization Lever Library (structured reference — NOT a score)

`engine/transition/levers.py` + `data/transition/lever_library.json`. Approach adapted from **BSR's Decarbonization Lever Library**: for each entity, map its sector to the specific decarbonization levers available to it, positioned by **value-chain stage** — upstream (purchased inputs/suppliers), own operations (Scope 1+2), downstream (products/customers/use-phase) — then overlay the entity's own **transition plan** to expose coverage and gaps.

- **29 levers in 5 domains** (Electricity & Energy, Transport, Industry, Buildings, FLAG & Water). Each lever carries: indicative abatement-cost band (\$/tCO₂e), global mitigation potential (high/med/low), commercial maturity (mature → frontier), 2050 role, key dependencies, a **nature & people** ("just transition") note at each value-chain stage, applicable sectors, and sources.
- **sector_lever_map** links all 30 taxonomy sectors to their levers, each tagged with value-chain position and relevance (primary/secondary) plus a rationale.
- **Plan overlay** (`overlay_plan`) is a **manual analyst input** — the analyst ticks which levers are in the plan — and returns a *factual* count (primary levers covered / total) and a gap list. It deliberately emits **no synthetic readiness score** (BSR "do-not-automate" boundary): the value is in surfacing which core abatement routes are unaddressed, not in rating the plan.

- **Data discipline:** the *mapping and value-chain logic* are carried over from BSR; the underlying cost/maturity/potential figures are **refreshed** indicative screening ranges (IEA NZE 2023, WEO 2024; IPCC AR6 WG3; IRENA 2024; Lazard 2025; Mission Possible Partnership) — not lifted from the original report.

Surfaced in the app as page 11 **Levers** (entity lever map + plan overlay + full-library reference, CSV export).

10. Data provenance & vintages

Component	Source	Vintage
Carbon prices (L1)	NGFS Phase V REMIND-MAGPIE 3.3-4.8 (IIASA), published Nov 2024	US\$2010 series ×1.18 (US GDP deflator) → US\$2020
Pass-through (L1)	Sijm 2012; Fabra & Reguant 2014; Cludius 2020	sector medians
Learning rates (L2)	Way et al. 2022; IRENA	2022–23
LCOE base costs (L2)	IRENA RPGC 2024; Lazard LCOE+ 2025 v18	2024–25
I-O matrix (L3)	EXIOBASE-3 (Stadler et al. 2018), IOT 2019	2019
CCEXposure + z-base (L4)	Sautner et al. JoF 2023 Tables 1 & 4	2002–2019, 10k firms
Sector pathways	NGFS / IEA WEO 2023	2023
MACC	IPCC AR6 WG3; IEA roadmaps	2022–23

11. Calibration — worked examples (default settings)

- **Coal plant** (\$500M, 2.5 MtCO₂): Net-Zero-2050 cumulative impairment **\$266.1M** — now reported as **53% recognised** stranded (= \$266M ÷ \$500M), against a **98% strandable ceiling** (crossover 2025). Current Policies ≈ \$33M.
- **Oil refinery** (\$2B, 12.8 MtCO₂): stranding triggered by **demand collapse** (crossover 2039), not cost crossover; cumulative impairment ≈ \$1.36B.
- **Office** (\$50M, 1k tCO₂): no stranding; small net cost / reputational uplift.

Impairment dollars are unchanged by the upgrades (abatement/priced default to legacy; anchor sectors hold slope 0.20). What changed at the external-review stage: the **% stranded label** now reports the recognised figure (53%) not the ceiling (98%); L1 cash-flow cost is lower where Scope 2/3 shift to L3 under the new **auto** default; and L3 is now materially larger after the emission-intensity scale correction.

12. Known limitations (challenge these)

- **Pass-through** is a static sector median; firm-specific values vary with market power and trade exposure. Override per asset.

- **CCExposure** sector exposures are anchored to published SIC-industry means, estimated for sectors not in Table 4; the licensed firm-level feed is the fix.
- **MRIO twin sectors** (`road_transport_ev` , `real_estate_residential`) are supply-split from their siblings (EXIOBASE has no separate EV / residential-RE industry).
- **Cascade 0 / contagion** are judgment parameters; the linear result is the default.
- **ITR / MACC / peer benchmark** are screening approximations aligned to SBTi/PACTA/PCAF/AR6 in spirit, not certified implementations.
- **Carbon-price vintage** is NGFS Phase V (Nov 2024); refresh for disclosure.
- Emissions decarbonise only if a target is set; without one, Scope 1+2 are held flat.
- **Structural simplifications (Round-1 U3)**: (i) IAM shadow prices are treated as *realised* carbon prices — a first-best assumption that overstates cost where policy underdelivers; (ii) the default world 20×20 matrix carries I-O aggregation bias; (iii) three price bands (advanced/emerging/RoW) proxy NGFS's ~12 model regions (the 20×49 MRIO mode relaxes this).
- **Endogenous-default cascade (Round-1 U2)** rests on Reisch et al. 2025, an un-peer-reviewed preprint — research-grade only; default-off and excluded from the disclosure export.
- **Firm-level roll-up (diagnostic fix #4)** is an independent sum of business lines — no group correlation, cross-subsidy, shared capital, or single optimised group transition plan.
- **io_proxy sectors (diagnostic fix #5/#6)**: the 7 sectors added after the 20×20 matrix (apparel, consumer goods, financial services, healthcare, professional services, telecom, metals mining) borrow an existing matrix row for Layer-3 propagation; a native row needs an I-O rebuild.
- **intermediate_input_share (diagnostic fix #1)** is a sector-median value-added split; firm-level COGS/input structure would sharpen L3.

Appendix A — Sector taxonomy (30 sectors)

Key · label · incumbent → challenger · fossil-dependent · emission intensity (tCO₂/M\$) · intermediate-input share · io_proxy (where applicable). See `data/transition/sector_taxonomy.json`. The **20 core** sectors span power (coal/gas/renewable), oil & gas (upstream/refining/distribution), heavy industry (steel/cement/chemicals/aluminium), transport (road ICE/EV, aviation, shipping), real estate (commercial/residential), agriculture, data centres, general manufacturing, and services. **+7 BSR lever-library sectors (diagnostic fix #5/#6)** added via `io_proxy`: apparel & textiles, consumer goods, financial services, healthcare, professional services, telecommunications, and metals & mining (a transition **beneficiary** — demand grows, no product-demand stranding). **+1 industrial & construction equipment (stress-test fix #2)**, also via `io_proxy` — a diesel-machinery maker (`diesel_machinery` → `electric_machinery`) whose dominant risk is the **use-phase** of its sold products (see \$4.5b), not its own operations. **+2 forest-products sectors (Mercer live test)**, via `io_proxy`: **pulp & paper** (biomass-powered low-carbon producer; not a stranding risk) and **solid wood / mass timber** (a transition **beneficiary** — mass timber displaces steel/cement, demand grows). Both carry a `biogenic_scope3_fraction` that nets biogenic wood fibre out of the report-anchored L3 (\$5.2b), so a pulp mill's fibre supply is not priced at the fossil carbon price.

Appendix B — Pass-through coefficients

See `data/transition/sector_pass_through.json` (pass-through, demand elasticity, market structure, trade exposure, source).

Appendix C — Learning rates & LCOE

See `data/transition/learning_curves.json`. Power LCOE (USD/MWh): coal 118, gas 76, solar 43, onshore wind 34, storage 170 (IRENA 2024 / Lazard 2025).

Appendix D — NGFS carbon prices

See `data/transition/carbon_prices_ngfs.json`. Real REMIND-MAgPIE, three bands, 2025–2050.

Appendix E — Layer-4 z-base & elasticities

Pooled distribution (Sautner Table 1, $\times 10^3$): opp 0.391/1.344 · reg 0.049/0.264 · phy 0.013/0.103 · total 0.943/2.443. Per-SD elasticities (bps): equity $18 \cdot z_{\text{reg}} + 9 \cdot z_{\text{phy}} - 25 \cdot z_{\text{opp}}$ (downside premium + opportunity discount); credit $12 \cdot z_{\text{reg}} + 6 \cdot z_{\text{phy}}$; revenue $35 \cdot z_{\text{opp}} - 25 \cdot z_{\text{reg}}$.

Appendix F — MACC measures

See `data/transition/macc.json`. Per-sector measures with marginal cost (USD/tCO₂) and abatement potential (share of Scope 1+2); AR6 WG3 / IEA-informed.

Appendix G — References

NGFS Phase V (IIASA) · Sijm 2012 · Fabra & Reguant 2014 · Cludius 2020 · **Farmer & Lafond, Research Policy 2016** · **Way, Ives, Mealy & Farmer, Joule 2022** · Lafond et al. TFSC 2018 · IRENA RPGC 2024 · Lazard LCOE+ 2025 v18 · BloombergNEF 2024 · Ziegler & Trancik 2021 · Stadler et al. (EXIOBASE) 2018 · Acemoglu et al. 2012 · Papageorgiou et al. 2017 · Reisch et al. 2025 (arXiv:2503.10644) · Sautner, van Lent, Vilkov, Zhang (JoF 2023; Mgmt Sci 2023) · Bolton & Kacperczyk (JFE 2021) · Zerbib (Rev. Finance 2019) · BSR *Decarbonization Lever Library* (Nov 2025) · IPCC AR6 WG3 · PCAF 2022 · SBTi · PACTA.